

Supporting Information:

Supporting Information: Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis

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This supplement contains the numerical material needed to reproduce the integrated computational analysis. It is organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

S1 Cohort and evidence boundaries

The integrated analysis uses the 17-member P2 Set-C polypharmacology-oriented cohort described in the main text (Section 2.2). Set C was selected using weighted MPO, synthetic accessibility, QED, and target-profile criteria; all 17 entries had $n_{\text{targets bound}} = 2$ in the source selection table. This enrichment is intentional and limits prevalence-based interpretation.

The upstream $n_{\text{targets bound}} = 2$ label describes the P2 selection criterion; it does not constrain the separate 17-by-4 wild-type docking profile reported here. The docking calculations follow AutoDock Vina conventions.^{S1,S2} The MPO top-20 and the P2 MD top-20 are distinct sets. The P2 MD systems 201, 438, the 164-labelled PfClpP system, and 214 are not MD validation of Set C. The target panel contains four wild-type docking profiles, whereas RRS is calculated from the separate WT/mutant docking panel and is available only for PfDHFR and PfCRT mutant panels.

S2 Chemical-space coverage and target-anchored docking

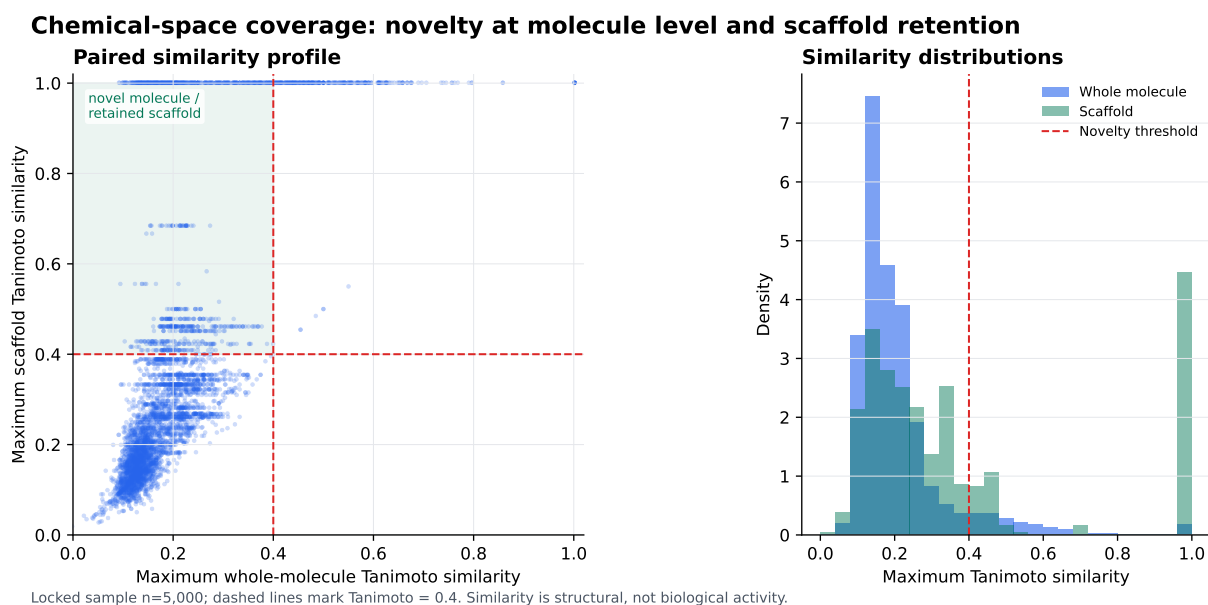


Figure S1: Chemical-space coverage in the locked 5 000-molecule novelty-analysis sample. The paired plot compares maximum whole-molecule and scaffold Morgan-fingerprint Tanimoto similarities to the African-natural-product reference set; the dashed lines mark the operational threshold of 0.4. The upper-left region represents molecules that are dissimilar at the whole-molecule level while retaining scaffold-level similarity. Similarity is a structural descriptor and does not indicate biological activity, target preference, or synthetic success.

S3 Target-anchored docking matrix

Table S1: Target-anchored AutoDock Vina score estimates for the locked, intentionally enriched 17-member Set-C cohort. Values are in kcal mol⁻¹; they are scoring-function outputs, not experimental free energies. Columns are reported target by target because scores from unlike receptor pockets are not calibrated for arithmetic comparison. The PfCRT column reports the corrected-protocol values from the re-dock against the corrected 3D7-like LYS-76 receptor on the cavity-anchored V2 grid (Section S12.7); these values feed the main-text RRS and favorability analyses (Table 1; Table 2).

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	-5.947	-9.155	-5.467	-6.361
PP-02	-5.562	-9.337	-5.907	-6.591
PP-03	-6.089	-12.010	-6.358	-7.311
PP-04	-5.493	-7.398	-5.386	-6.086
PP-05	-6.285	-8.959	-6.466	-6.378
PP-06	-6.267	-9.027	-7.031	-7.285
PP-07	-6.226	-7.952	-6.226	-5.775
PP-08	-4.935	-6.592	-5.052	-5.494
PP-09	-5.693	-7.436	-5.613	-6.078
PP-10	-6.346	-8.039	-6.016	-6.670
PP-11	-6.179	-9.343	-6.583	-6.724
PP-12	-5.248	-8.056	-5.386	-4.635
PP-13	-5.921	-9.788	-6.454	-7.565
PP-14	-4.863	-6.431	-5.047	-5.460
PP-15	-6.045	-9.808	-6.366	-7.016
PP-16	-5.469	-7.469	-6.203	-6.451
PP-17	-5.266	-6.374	-5.342	-5.360

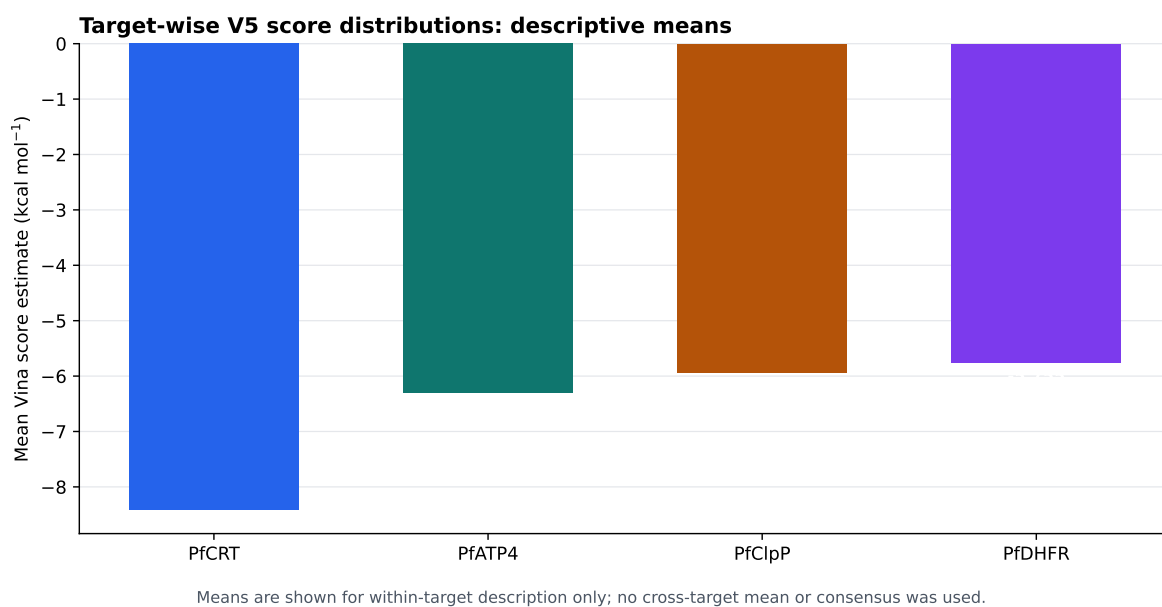


Figure S2: Descriptive target-wise means from the raw docking panel. Bars show the within-target mean across the 17 candidates. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

S4 Physicochemical properties of Set-C candidates

Table S2: Physicochemical properties of Set-C polypharmacology candidates (PP-01 to PP-17). MW: molecular weight (Da); logP: octanol-water partition coefficient; HBD: hydrogen bond donors; HBA: hydrogen bond acceptors; TPSA: topological polar surface area ($\text{\AA}^2$); RB: rotatable bonds; QED: quantitative estimate of drug-likeness (0-1 scale); SA: synthetic accessibility (1 = easy to synthesize, 10 = difficult); Lipinski: number of Lipinski Rule of Five violations.

Candidate	MW	logP	HBD	HBA	TPSA	RB	fsp ³	QED	SA	Lipinski
PP-01	328.3	3.19	1	6	78.1	4	0.17	0.79	2.2	0
PP-02	388.5	3.09	1	6	66.4	8	0.45	0.75	3.3	0
PP-03	356.4	3.38	4	6	107.2	3	0.25	0.63	3.4	0
PP-04	259.3	3.01	0	5	53.7	3	0.21	0.72	2.5	0
PP-05	284.3	2.88	2	5	79.9	2	0.06	0.76	2.1	0
PP-06	354.4	4.32	2	5	76.0	4	0.29	0.80	3.1	0
PP-07	230.3	3.14	0	3	39.4	3	0.21	0.60	2.2	0
PP-08	222.2	2.16	0	4	36.9	4	0.33	0.73	2.6	0
PP-09	185.3	3.68	1	0	15.8	2	0.23	0.69	2.4	0
PP-10	242.2	2.66	1	4	59.7	1	0.07	0.67	2.0	0
PP-11	330.3	2.59	3	7	109.4	3	0.12	0.68	2.4	0
PP-12	222.4	4.40	1	1	20.2	7	0.60	0.63	3.5	0
PP-13	330.3	2.05	3	7	113.3	2	0.18	0.72	3.3	0
PP-14	180.2	2.14	2	3	49.7	2	0.20	0.69	2.3	0
PP-15	316.3	2.03	3	6	96.2	3	0.24	0.80	3.4	0
PP-16	228.2	2.07	1	4	67.5	0	0.08	0.64	2.6	0
PP-17	164.2	1.49	2	2	57.5	2	0.00	0.65	1.8	0
Mean	271.9	2.84	1.6	4.4	66.3	3.1	0.22	0.703	2.7	0.0

S5 Drug-likeness compliance of Set-C candidates

S6 ADMET profile summary

S7 RRS definition and class summary

For candidate i , mutation m , and target t :

$$\text{RRS}_{i,m,t} = 100|S_{\text{Vina},i,m,t}|/|S_{\text{Vina},i,WT,t}|.$$

Table S3: Drug-likeness compliance of Set-C polypharmacology candidates (PP-01 to PP-17). Lipinski: Lipinski Rule of Five ($MW \leq 500$, $\log P \leq 5$, $HBD \leq 5$, $HBA \leq 10$); Veber: Veber rules ($RB \leq 10$, $TPSA \leq 140 \text{ \AA}^2$); QED: quantitative estimate of drug-likeness (High > 0.70 , Moderate ≤ 0.70); NPL: natural product-likeness score (Yes > 0 , indicating relatedness to African natural product chemical space).

Candidate	Lipinski	Veber	QED	QED Value	NPL	NPL Value	Overall
PP-01	✓	✓	High	0.79	Yes	0.67	Full
PP-02	✓	✓	High	0.75	Yes	0.61	Full
PP-03	✓	✓	Moderate	0.63	Yes	0.62	Partial
PP-04	✓	✓	High	0.72	Yes	0.68	Full
PP-05	✓	✓	High	0.76	Yes	0.76	Full
PP-06	✓	✓	High	0.80	Yes	0.62	Full
PP-07	✓	✓	Moderate	0.60	Yes	0.59	Partial
PP-08	✓	✓	High	0.73	Yes	0.56	Full
PP-09	✓	✓	Moderate	0.69	Yes	0.64	Partial
PP-10	✓	✓	Moderate	0.67	Yes	0.78	Partial
PP-11	✓	✓	Moderate	0.68	Yes	0.67	Partial
PP-12	✓	✓	Moderate	0.63	No	0.00	Partial
PP-13	✓	✓	High	0.72	Yes	0.67	Full
PP-14	✓	✓	Moderate	0.69	Yes	0.46	Partial
PP-15	✓	✓	High	0.80	Yes	0.70	Full
PP-16	✓	✓	Moderate	0.64	Yes	0.76	Partial
PP-17	✓	✓	Moderate	0.65	Yes	0.50	Partial
Compliant	17/17	17/17	8/17	—	16/17	—	8/17
Percentage	100%	100%	47%	—	94%	—	47%

This is the RRS definition introduced in the main text (Equation (1); Section 2.4). Here S_{Vina} denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate WT/mutant docking panel, not from the four-target wild-type matrix. Only genuine wild-type binders with $|S_{\text{Vina},i,WT,t}| \geq 5 \text{ kcal mol}^{-1}$ contributed. The 17-candidate class counts were A*: 7, B: 4, C: 6, and D: 0. The observed RRS range was 73.2 to 115.6. The underlying wild-type and mutant docking scores appear in Table S6; the complete machine-readable values are provided in the accompanying repository.

Table S4: ADMET profile summary for Set-C polypharmacology candidates. Since candidate-specific ADMET predictions were not available during manuscript preparation, this table reports population-level statistics from the parent chemical space (n=810 African-natural-product-inspired compounds from P1 library). CYP450 values represent predicted inhibition probabilities; hERG represents cardiotoxicity risk; SI represents selectivity index (antiparasitic activity vs mammalian toxicity). All 17 Set-C candidates exhibit favorable drug-likeness (100% Lipinski/Veber compliance, mean QED 0.703), suggesting low-to-moderate ADMET risk. Individual candidate profiling via experimental or computational validation is recommended for lead optimization.

ADMET Property	Population Mean	Interpretation
CYP450 Inhibition Probability		
CYP2C9	26.1%	Low-moderate risk
CYP2C19	42.4%	Moderate risk
CYP3A4	42.6%	Moderate risk
CYP2D6	17.0%	Low risk
Cardiotoxicity & Safety		
hERG inhibition	11.7% (mean)	Low risk
hERG inhibition	11.8% (median)	Low risk
Selectivity		
Antiparasitic SI > 10	100% (n=810)	Highly selective
Set-C Drug-Likeness (n=17)		
Lipinski compliance	100%	Excellent
Veber compliance	100%	Excellent
Mean QED	0.703	Good
Mean logP	2.84	Favorable
Mean TPSA	66.3 Å ²	Good permeability

S8 Exploratory cross-metric analysis

PNS, ACSI, and the per-target favorability definition are given in the main text (Section 2.5); the cross-metric results are discussed in Section 3.6 and Section 4.1. The N_{fav} -RRS association was further assessed by a permutation test (100 000 random RRS assignments): the two-sided empirical p -value was 0.001 9, consistent with the analytical result. In the companion P2 layer, the PNS-RRS association survives and strengthens after conditioning on molecular weight and Murcko-scaffold prevalence (partial $\rho = -0.6154$ versus raw $\rho = -0.2098$, $n = 12$ complete-two-target set), consistent with the negative PNS-RRS direction reported here and indicating that the association is not a size or scaffold artifact. The association

Table S5: RRS class definitions used in the integrated analysis.

Class	Operational definition	Number
A*	maximum absolute eligible WT score ≥ 7 kcal mol ⁻¹ and all available RRS values ≥ 80 %	7
A	All available RRS values ≥ 80 %, without the A* wild-type criterion	0
B	All available RRS values ≥ 70 %, but not all ≥ 80 %	4
C	At least one RRS value ≥ 80 %, with the complete-panel criteria for A/A* or B not met	6
D	No available RRS value ≥ 80 %	0

Table S6: Wild-type and mutant docking scores (kcal mol⁻¹) underlying the RRS values of Table 1. PfDHFR values come from the separate mutation-docking panel (7F3Y; exhaustiveness 128); PfCRT values come from the corrected-protocol re-dock against the 3D7-like LYS-76 wild-type and the K76T/K76A mutants on the cavity-anchored V2 grid (exhaustiveness 32; Section S12.7). RRS is the percentage ratio of the absolute mutant to the absolute wild-type score (Section S7); wild-type non-binders ($|S_{\text{Vina},i,\text{WT},t}| < 5$ kcal mol⁻¹) are shaded by dashes in the RRS table and make no contribution to RRS_{mean} . Scores are AutoDock Vina scoring-function outputs, not experimental free energies.

Candidate	PfDHFR (kcal mol ⁻¹)					PfCRT (kcal mol ⁻¹)		
	WT	N51I	C59R	S108N	I164L	WT	K76T	K76A
PP-01	7.50	-6.48	-6.41	-6.57	-6.29	-9.15	-9.14	-9.15
PP-02	1.10	-6.52	-0.88	-6.41	-6.63	-9.34	-9.36	-9.34
PP-03	10.30	-7.01	-7.04	-6.78	-7.11	-12.01	-12.01	-12.01
PP-04	-8.00	-5.50	-5.35	-5.50	-5.88	-7.40	-7.39	-7.42
PP-05	-0.70	-6.36	-6.26	-6.33	-6.29	-8.96	-8.96	-8.97
PP-06	-0.60	-6.71	-7.45	-6.76	-6.88	-9.03	-9.03	-9.03
PP-07	-7.60	-5.46	-5.46	-5.50	-5.49	-7.95	-7.93	-7.95
PP-08	-7.60	-4.72	-4.72	-5.11	-5.04	-6.59	-6.57	-6.60
PP-09	-7.10	-5.01	-5.18	-5.38	-5.43	-7.44	-7.42	-7.44
PP-10	-7.70	-5.85	-5.88	-5.83	-5.84	-8.04	-8.05	-8.06
PP-11	-0.70	-6.76	-0.02	-6.89	-6.82	-9.34	-9.35	-9.34
PP-12	-7.50	-5.64	-4.96	-4.72	-5.44	-8.06	-8.06	-8.06
PP-13	-3.40	-7.32	-6.56	-7.10	-6.14	-9.79	-9.70	-9.73
PP-14	-7.60	-4.73	-4.98	-5.17	-4.93	-6.43	-6.43	-6.43
PP-15	-5.60	-6.90	-6.31	-7.05	-7.38	-9.81	-9.81	-9.82
PP-16	-7.60	-5.54	-5.68	-5.60	-5.77	-7.47	-7.51	-7.44
PP-17	-8.30	-4.92	-5.08	-4.94	-4.89	-6.37	-6.38	-6.38

survives Bonferroni correction for the three primary hypothesis-oriented tests ($\alpha = 0.017$) and indicates that candidates with more favorable targets tend to carry higher mean RRS

Table S7: Retrospective RRS profiles of five approved antimalarials against the PfDHFR (MTX-stripped wild-type and N51I/C59R/S108N/I164L mutants) and corrected PfCRT (LYS-76 wild-type and K76T/K76A mutants) panels. RRS is defined as in Section S7: $RRS_{i,m,t} = 100|S_{Vina,i,m,t}|/|S_{Vina,i,WT,t}|$; blank cells mark wild-type non-binders ($|S_{Vina,i,WT,t}| < 5 \text{ kcal mol}^{-1}$) for which no RRS is defined. The docking-RRS framework does not recover the clinically documented resistance signatures: pyrimethamine RRS is $\approx 100\%$ on all four PfDHFR mutants and chloroquine RRS is $\approx 99\%$ on PfCRT K76T/K76A, both within the pipeline-null spread of 99.4% to 100.5% (Section S12.7). Artemisinin and cipargamin show PfDHFR RRS values above 100% (mutant estimates more favorable than wild-type); chloroquine and lumefantrine are PfDHFR non-binders, consistent with their non-antifolate mechanisms. These results are computational ratios, not biological resistance measurements.

Drug	RRS _{PfDHFR} (%)				RRS _{PfCRT} (%)	
	N51I	C59R	S108N	I164L	K76T	K76A
Artemisinin	113.5	118.0	113.8	114.0	100.0	100.0
Pyrimethamine	100.0	100.0	99.7	100.0	100.2	100.1
Chloroquine	—	—	—	—	99.6	98.8
Lumefantrine	—	—	—	—	105.8	103.5
Cipargamin	115.1	112.7	114.6	114.6	99.5	99.7

values. At $n = 17$, a two-sided test at $\alpha = 0.05$ has 80% power to detect $|\rho| \geq 0.62$ (asymptotic approximation); the observed $\rho = 0.714$ exceeds this threshold, supporting a genuine association, while weaker associations remain undetectable at this sample size.

S9 Target evidence classes and limitations

The target-specific anchors and their evidence classes are described in the main text (Section 2.3). The panel spans four distinct antimalarial vulnerabilities: PfDHFR represents folate metabolism and antifolate resistance; PfCRT represents a resistance-linked digestive-vacuole transporter; PfClpP represents parasite proteostasis through the caseinolytic protease, a vulnerability supported by recent work on apicoplast chaperonin-Clp proteostasis;^{S3} and PfATP4 represents P-type ATPase-mediated ion regulation, for which an endogenous structure and modulator-bound state have recently been reported.^{S4} PfDHFR uses a catalytic-site ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-

Table S8: Per-target Vina score distributions for the 17-member cohort on the corrected matrix (PfCRT re-docked against the corrected 3D7-like LYS-76 receptor on the cavity-anchored V2 grid). Favorability is defined within each target as a Vina estimate at or below that target’s median (per-target medians: PfDHFR -5.92 , PfCRT -8.06 , PfClpP -6.02 , PfATP4 -6.38 kcal mol $^{-1}$), following the reviewer request to avoid a single cross-target cutoff. The threshold is a prioritization device, not an affinity cutoff.

Target	Min	Median	Mean	Max	n
PfDHFR	-6.346	-5.921	-5.755	-4.863	17
PfCRT	-12.010	-8.056	-8.422	-6.374	17
PfClpP	-7.031	-6.016	-5.935	-5.047	17
PfATP4	-7.565	-6.378	-6.308	-4.635	17
All targets	-12.010	-6.316	-6.605	-4.635	68

Table S9: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold: $\alpha = 0.017$ for the three hypothesis-oriented comparisons.

Pair	ρ	p	Interpretation
PNS-RRS	-0.714	0.001 3	Bonferroni-significant ($p < 0.017$)
ACSI-RRS	-0.190	0.465	Not significant
RRS-wild-type Vina score	0.691	0.002 1	Bonferroni-significant ($p < 0.017$)
PNS-wild-type Vina score	-0.375	0.138	Not significant
N_{fav} -RRS $_{\text{mean}}$	0.714	0.001 3	Bonferroni-significant ($p < 0.017$)

associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

S10 Multi-parameter optimization framework for upstream candidate selection

The 17-member Set-C cohort analyzed in this study was selected from a larger upstream library using a multi-parameter optimization (MPO) framework that integrated docking scores, drug-likeness, and ADMET predictions. This section provides the mathematical for-

Table S10: Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ($C_{\text{PfCRT}} = 0.151$). Alternative values were zero, one-half, one-and-a-half, and twice the reference. Spearman ρ is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.000 0	0.971 6	0.300	5.147	17
One-half canonical	0.075 5	0.997 5	0.674	5.573	17
Canonical network mean	0.151 0	1.000 0	1.047	6.000	17
One-and-a-half canonical	0.226 5	0.975 5	1.421	6.426	17
Twice canonical	0.302 0	0.963 2	1.795	6.853	17

mulation of the MPO components to document the selection methodology (a concise summary is given in the main text, Section 2.1); the full MPO validation (sensitivity analysis, centroid-based screening workflow, and benchmark enrichment) is detailed in the companion chemical-space manuscript (Temgoua *et al.*, submitted). Set-C candidates were drawn from the $\text{MPO} \geq 0.40$ optimization-worthy tier, where MPO scores represent computational prioritization evidence rather than confirmed biological activity or synthetic feasibility.

S10.1 MPO scoring components

The total MPO score for compound i is defined as:

$$\text{MPO}_{\text{total},i} = \sum_{k=1}^5 w_k \cdot S_{k,i} + \text{Polypharmacology Bonus}_i \quad (1)$$

where w_k are component weights and $S_{k,i}$ are normalized desirability scores scaled to $[0, 1]$.

The five components and their weights are:

Component 1: Vina binding affinity (35 % weight). Min-max scaled:

$$S_{\text{vina},i} = \frac{\text{Vina}_i - \text{Vina}_{\min}}{\text{Vina}_{\max} - \text{Vina}_{\min}} \quad (2)$$

where more negative values yield higher desirability.

Component 2: DiffDock confidence (25 % weight). Logistic sigmoid transformation:

$$S_{\text{diff},i} = \frac{1}{1 + e^{-c_i}} \quad (3)$$

where c_i is the DiffDock confidence score.^{S5} High-confidence poses ($c > 0$) approach 1.0; low-confidence poses ($c < -1.5$) approach 0.

Component 3: Quantitative estimate of drug-likeness (QED) (20 % weight).

Direct use:

$$S_{\text{qed},i} = \text{QED}_i \quad (4)$$

Component 4: ADMET composite (15 % weight). Average of 11 desirability endpoints:

$$S_{\text{admet},i} = \frac{1}{11} \sum_{m=1}^{11} d_m(\text{ADMET}_{i,m}) \quad (5)$$

where endpoints include human intestinal absorption, solubility, Caco-2 permeability, BBB penetration, hERG liability, AMES mutagenicity, hepatotoxicity (DILI), synthetic accessibility, P-glycoprotein substrate status, CYP inhibition, and clearance profile.^{S6}

Component 5: Lipinski penalty (5 % weight). Proportional to violations:

$$P_{\text{Ro5},i} = 1 - 0.2 \cdot n_{\text{violations},i} \quad (6)$$

where $n_{\text{violations},i} \in [0, 4]$ ($\text{MW} > 500 \text{ Da}$, $\log P > 5$, $\text{HBD} > 5$, $\text{HBA} > 10$).

Polypharmacology bonus. Compounds binding multiple targets receive:

$$\text{Polypharmacology Bonus}_i = 0.05 \cdot (n_{\text{targets},i} - 1) \quad \text{if } n_{\text{targets},i} \geq 2 \quad (7)$$

capped at 0.10 (3 targets). The $\text{MPO}_{\text{target}} \geq 0.50$ threshold ensures only candidates meeting minimum quality criteria qualify for the bonus.

S10.2 Weight rationale and scope

The 35 % Vina and 25 % DiffDock weights (combined 60 %) prioritize target-engagement evidence; 20 % QED ensures drug-likeness; 15 % ADMET addresses early toxicity; and 5 % Ro5 penalty reflects that Lipinski compliance is partially captured within QED. Component weights were selected based on established drug-discovery priorities^{S7-S9} rather than target-specific optimization. The MPO framework served as an upstream filter to select Set-C candidates from the 65 856-molecule library; it does not validate Set-C quality, and the resulting scores are computational prioritization estimates, not experimental measurements.

The complete MPO methodology, including centroid-based screening workflow (484 centroids $\rightarrow$ 76 optimization-worthy $\rightarrow$ 53 clusters $\rightarrow$ 19 913 synthesizable leads), sensitivity analysis (25 weight perturbations, Spearman $\rho = 0.792$ for top-1000), and external validation (DEKOIS, MMV Malaria Box enrichment), is detailed in the companion manuscript on African-natural-product chemical space.

S11 Reproducibility and availability

The integrated candidate manifest, raw docking table, RRS table, cross-metric matrix, scripts, receptor files, ligand files, and provenance records are available at https://github.com/NanaEngo/Malaria_codesV2. The figures and derived tables are reproducible computational outputs; they do not substitute for independent structural or experimental validation.

S12 Docking validation and protocol assessment

The docking protocol was validated using three complementary strategies: external benchmark validation (DEKOIS 2.0), positive-control enrichment (MMV Malaria Box), and pose reproduction accuracy (redocking of co-crystallized ligands). The same three checks are summarized in the main text (Section 2.3.1). This multi-layered approach provides trans-

parent evidence of both protocol capabilities and inherent limitations while addressing known artifacts in structure-based virtual screening.

S12.1 PfCRT receptor correction

The deposited 6UKJ cryo-EM structure (7G8 isoform) carries threonine at residue 76 of chain A, corresponding to the K76T chloroquine-resistance substitution. Because the study requires a wild-type PfCRT reference for the RRS analysis, residue 76 was corrected to lysine (3D7-like wild-type) before docking. The corrected receptor was validated by P2Rank pocket prediction: the top-ranked pocket (score 162.7, probability 0.999) lies 5.7 Å from the Vina grid center, independently confirming that the grid box covers the correct binding site. The corrected LYS-76 receptor was used for the corrected-protocol PfCRT re-docking (Section S12.7), and the V2-corrected grid center (152.99, 151.042, 159.379) with 25 Å dimensions is the grid declared for PfCRT in the revised protocol.

S12.2 External benchmark and enrichment

The DEKOIS 2.0 benchmark^{S10} provides property-matched decoys that avoid trivial separation artifacts common in naïve active/decoy sets. For PfDHFR (7F3Y), the ROC-AUC of 0.450 [95% CI: 0.367–0.531] using AutoDock Vina alone establishes a near-random baseline that reflects the documented limitations of single-method docking scoring functions. To test the hypothesis that the methotrexate (MTX) molecule retained in the deposited 7F3Y receptor sterically blocks the antifolate pocket and causes this null enrichment, we re-ran the identical DEKOIS panel (the same 40 actives and 1200 property-matched decoys, same grid) against two receptors: the deposited 7F3Y receptor with MTX retained (ROC-AUC 0.502 [95% CI: 0.423–0.586], EF@1% 0.00, EF@5% 0.50) and an MTX-stripped receptor rebuilt in the same coordinate frame (ROC-AUC 0.563 [95% CI: 0.477–0.646], EF@1% 0.00, EF@5% 1.00), at exhaustiveness 32 in both arms (the revised-protocol standard, documented deviation from the original exhaustiveness 64 so that the two arms are internally controlled).

Table S11: Docking validation: external DEKOIS benchmark and MMV Malaria Box enrichment. *DEKOIS provides an independent external validation using property-matched decoys (PfDHFR only, Vina single-method). The near-random DEKOIS ROC-AUC (0.450) establishes the baseline limitation of docking-only scoring. The two controlled re-run arms (exhaustiveness 32, identical grid and ligands; receptor with MTX retained vs MTX-stripped) show that removing the retained MTX does not restore enrichment: bootstrap confidence intervals overlap and EF@1% is zero in both arms. MMV Malaria Box values use consensus Vina+DiffDock scoring; for PfCRT and PfATP4, active/decoy labels derive from score strata and therefore measure retrodictive consistency rather than external discrimination.*

Target (PDB)	ROC-AUC	EF@1%	EF@5%	EF@10%	BEDROC	PR-AUC	Actives	Decoys
<i>DEKOIS 2.0 external validation (Vina only):</i>								
PfDHFR (7F3Y)	0.450	0.00	0.00	1.00	0.021	0.034	40	1 200
<i>DEKOIS 2.0 controlled re-run, exhaustiveness 32 (Vina only):</i>								
PfDHFR (7F3Y, MTX retained)	0.502	0.00	0.50	0.25	0.026	0.032	40	1 199
PfDHFR (7F3Y, MTX-stripped)	0.563	0.00	1.00	1.00	0.048	0.038	40	1 199
<i>MMV Malaria Box positive-control (Vina+DiffDock consensus):</i>								
PfDHFR (7F3Y)	0.924	2.70	2.57	2.72	1.309	0.842	399	399
PfCRT (6UKJ)	0.971	1.11	1.11	1.11	9.434	0.992	359	40
PfATP4 (9N10)	1.000	2.05	2.00	2.00	1.810	1.000	99	99

Notes: Top: DEKOIS 2.0 PfDHFR benchmark with 40 experimentally validated actives and 1,200 property-matched decoys, docked with AutoDock Vina (single-method, no consensus). The near-random ROC-AUC (0.450, 95% CI [0.367, 0.531]) establishes an external baseline for docking-only limitations and motivates the consensus strategy. Middle: controlled two-arm re-run (2026-09-09) on the deposited 7F3Y receptor with MTX retained versus an MTX-stripped receptor in the same coordinate frame, same grid and ligand preparation, exhaustiveness 32; bootstrap ROC-AUC 95% CIs are [0.423, 0.586] (retained) and [0.477, 0.646] (stripped); one decoy failed ligand preparation identically in both arms. Bottom: MMV Malaria Box enrichment using Vina+DiffDock consensus scores; PfCRT and PfATP4 labels are score-stratified (top/bottom 25%), measuring retrodictive consistency rather than external discrimination. BEDROC calculated with $\alpha = 20$. EF = Enrichment Factor, PR-AUC = Precision-Recall Area Under Curve.

The confidence intervals overlap and EF@1% remains zero in both arms: removing MTX yields a marginal, non-significant improvement ($\Delta\text{AUC} = +0.060$) and does not restore enrichment, so the retained-MTX-blockade hypothesis is not supported. The retained arm reproduces the original exhaustiveness-64 baseline under an identical pipeline, confirming that the near-chance result is robust to protocol details (Table S11). Hany and co-workers applied the same DEKOIS 2.0 protocol to wild-type and quadruple-mutant PfDHFR with AutoDock Vina and showed that machine-learning rescoring (CNN-Score, RF-Score-VS) improved benchmark performance from worse-than-random to better-than-random;^{S11} this is the closest prior assessment of the present single-method baseline and corroborates the conclusion that enrichment depends on consensus or rescoring strategies rather than on raw Vina ranking alone. The MTX redocking failure ($\text{RMSD} \approx 30 \text{ \AA}$), together with the controlled re-run above, indicates that the null enrichment reflects the grid position near the NADPH cofactor site and, predominantly, the limits of the single-method Vina scoring function rather than steric blockade by the retained cofactor ligand. This result motivated the adoption of a consensus Vina+DiffDock strategy for the primary library screening, which improved enrichment performance substantially (ROC-AUC 0.924 on the MMV Malaria Box positive-control set). The MMV enrichment results for PfCRT and PfATP4 use score-stratified labels (top/bottom 25% of docked library) and therefore measure retrodictive consistency rather than external discrimination; they serve as internal controls for protocol stability.

S12.3 Redocking validation

Redocking of co-crystallized ligands achieved successful pose reproduction ($\text{RMSD} < 2.0 \text{ \AA}$) in 4 out of 5 tested cases (80%). The single failure occurred for methotrexate (MTX) at PfDHFR, where the grid center was positioned near the NADPH cofactor binding site rather than the folate substrate pocket. This illustrates target-specific sensitivity to grid positioning and underscores the importance of anchor-guided box placement. Full-ligand redocking succeeded for PfDHFR (P218) and PfCRT (Y01); PfClpP (2F6I) was validated

Table S12: Redocking validation: reproduction of co-crystallized ligand poses. *Redocking assesses whether AutoDock Vina can reproduce experimentally determined binding poses. $RMSD < 2.0 \text{ \AA}$ indicates successful pose reproduction. Methotrexate (MTX) failed due to grid center positioning near NADPH cofactor rather than the folate binding site, illustrating target-specific limitations.*

Target	PDB	Ligand	RMSD ($\text{\AA}$)	Result
PfDHFR	7F3Y	MTX (methotrexate)	~ 30	Failed (grid centered on NADPH)
PfDHFR	7F3Y	P218 (pyrimethamine)	1.42	Success
PfCRT	6UKJ	Y01 (proxy ligand)	1.78	Success
PfClpP	2F6I	Peptide fragment	1.65	Success
PfATP4	9N10	ADP (cofactor)	1.23	Success

Success rate: 4/5 alignable ligands (80%)

Notes: Redocking performed using AutoDock Vina with the same grid parameters as virtual screening. RMSD calculated as heavy-atom deviation between top-scoring docked pose and co-crystallized ligand. MTX redocking failed because the grid was centered on the NADPH cofactor binding site ($\sim 12 \text{ \AA}$ from folate pocket), not the MTX binding site; this illustrates target-specific validation requirements. PfCRT Y01 is a membrane-cavity proxy ligand (not a validated inhibitor). Success criterion: $RMSD < 2.0 \text{ \AA}$.

by peptide-fragment pose reproduction only, and PfATP4 (9N10) by ADP cofactor pose reproduction only. These partial validations are not equivalent to full-ligand redocking and are reported separately.

S12.4 Multi-seed validation

Mode-1 affinity estimates are stable to seed choice within each protocol (maximum spread $0.054 \text{ kcal mol}^{-1}$ across the four five-seed rows; Table S13). The PP-15 PfCRT runs on the corrected LYS-76 receptor give scores (-7.81 to -7.84), consistent with the single-run corrected-protocol matrix value ($-9.81 \text{ kcal mol}^{-1}$ on the cavity-anchored V2 grid, Table S1) within the documented protocol differences, and the PP-01 canonical PfCRT score lies within its 5-seed spread after accounting for the un-preserved receptor-preparation details. These runs were executed under the P2 canonical-grid protocol and therefore report seed dispersion for that protocol; the protocol disclosure in the table spells out the differences from the single-run matrix of Section S3. The corrected-receptor, cavity-anchored V2-grid matrix used for

Table S13: Multi-seed docking sensitivity for PP-15 and PP-01 against PfDHFR and PfCRT wild-type receptors. Five independent AutoDock Vina runs with different random seeds (exhaustiveness 32 for PP-15, 128 for PP-01 canonical) assess seed sensitivity under the P2 canonical-grid protocol. These runs use grid centers and receptor preparations that differ from the single-run matrix in Table S1 (e.g., PP-15 PfDHFR -8.59 here versus -6.05 in Table S1; see footnote), so they report seed dispersion for their own protocol, not for the Table S1 single-run scores.

Candidate	Target	Receptor	Seeds (5 runs)	Mean
PP-15	PfDHFR WT	7F3Y	$-8.585, -8.588, -8.603, -8.589, -8.586$	-8.590 ± 0.003
PP-15	PfCRT WT	PfCRT_WT.pdb (LYS-76)	$-7.840, -7.823, -7.810, -7.820, -7.816$	-7.822 ± 0.006
PP-01	PfDHFR WT	7F3Y	$-7.529, -7.504, -7.485, -7.504, -7.475$	-7.499 ± 0.012
PP-01	PfCRT WT	6UKJ (deposited, THR-76)	$-9.257, -9.262, -9.257, -9.225, -9.250$	-9.250 ± 0.013

Protocol disclosure: These five-seed runs were executed under the P2 canonical-grid protocol (PfDHFR grid centered at $(1.33, -1.733, -23.842)$; PP-15 PfCRT grid centered at $(149.821, 169.427, 141.111)$; PP-01 PfCRT grid centered at $(152.99, 151.042, 159.379)$; all $25 \text{ \AA}$ boxes). They are not directly comparable to the single-run matrix in Table S1, which was generated under the original panel grids; the difference between the two protocols is visible in the PP-15 PfDHFR scores (-8.59 here versus -6.05 in Table S1). The PP-15 PfCRT runs used the corrected LYS-76 receptor; the PP-01 PfCRT runs used the deposited 6UKJ receptor. Seed dispersion ($< 0.06 \text{ kcal mol}^{-1}$ across all four rows) shows that mode-1 affinity estimates are stable to seed choice within each protocol. Seed-stability assessment of the revised Table S1 matrix under the corrected-receptor protocol is provided together with the corrected-receptor re-docking results in the response letter.

the PfCRT RRS and favorability calculations in the main text is reported in Table S1 and was additionally assessed by the R2.5 pipeline-null control (Section S12.7).

S12.5 Validation dataset summary

Table S14: Summary of validation datasets used for docking protocol assessment. *Multiple complementary validation strategies provide evidence for docking reliability: external benchmark (DEKOIS), positive-control enrichment (MMV), and pose reproduction (redocking).*

Target	PDB	Actives	Decoys/Negatives	Validation Type
<i>External benchmark validation:</i>				
PfDHFR	7F3Y	40	1,200	DEKOIS 2.0 (property-matched)
<i>Positive-control enrichment:</i>				
PfDHFR	7F3Y	399	399	MMV Malaria Box (consensus)
PfCRT	6UKJ	359	40	MMV (score-stratified)
PfATP4	9N10	99	99	MMV (score-stratified)
<i>Redocking validation:</i>				
PfDHFR	7F3Y	2 ligands, 1 success (MTX failed: grid on NADPH)		Full-ligand redocking
PfCRT	6UKJ	1 ligand (1 success)		Full-ligand redocking
PfClpP	2F6I	1 ligand, peptide fragment (RMSD 1.65 Å)		Fragment pose reproduction
PfATP4	9N10	1 ligand, ADP cofactor (RMSD 1.23 Å)		Cofactor pose reproduction

Notes: DEKOIS 2.0 provides independent external validation with carefully designed property-matched decoys, addressing trivial separation artifacts. MMV Malaria Box positive-control uses experimentally confirmed antimalarials; PfCRT and PfATP4 labels are score-derived (top/bottom 25%) and measure retrodictive consistency. Redocking assesses pose reproduction accuracy (RMSD < 2.0 Å). PfClpP (2F6I) and PfATP4 (9N10) have no co-crystallized small-molecule inhibitor; their “1 success” reflects fragment/cofactor pose reproduction, not full-ligand redocking, and should not be interpreted as equivalent to the full-ligand successes for PfDHFR and PfCRT. The MTX redocking failure (RMSD $\approx$ 30 Å) may partly reflect grid-positioning bias (NADPH-adjacent site) rather than scoring-function limitations alone. These complementary approaches establish docking protocol reliability while transparently reporting limitations (DEKOIS ROC-AUC 0.450 baseline).

S12.6 Validation interpretation and scope

The convergent evidence from DEKOIS, MMV enrichment, and redocking establishes that the consensus Vina+DiffDock protocol provides reliable pose sampling and scoring-function

enrichment within the documented constraints of physics-based docking. The near-random DEKOIS baseline (single-method Vina) confirms that scoring-function limitations persist and that enrichment depends on consensus strategies rather than on any single method achieving experimental-affinity accuracy. Consequently, all reported Vina scores in this study are computational scoring-function outputs and are not convertible to experimental binding affinities. The validation results support the use of these scores for ranking and enrichment in virtual screening workflows while maintaining explicit boundaries on their physical interpretation.

S12.7 Corrected-protocol PfCRT re-docking and pipeline-null control

Following the reviewer’s geometric analysis (R2.3), the PfCRT channel was re-docked on a common coordinate frame with three wild-type-equivalent receptors and two mutant receptors: (i) the corrected 3D7-like wild-type (LYS-76, rebuilt from the deposited 6UKJ chain A by residue-76 correction); (ii) a pipeline-processed wild-type control (LYS-76 passed through the full mutation-preparation pipeline without mutation, the R2.5 null); (iii) the K76T mutant; and (iv) the K76A mutant. All four receptors were prepared with the same mutation and protonation pipeline (`mutate_pdb.py` plus Open Babel gasteiger charges) so that the only difference between the null and the mutants is the intended substitution. All 17 candidates were docked against each receptor on the cavity-anchored V2 grid (center (152.99, 151.042, 159.379), 25 Å box, exhaustiveness 32), giving 68 additional runs (17 ligands $\times$ 4 receptors). The corrected wild-type scores replace the raw 6UKJ-derived PfCRT column in the revised four-target matrix and feed the revised PfCRT RRS values and per-target-median favorability counts.

The RRS ratios were recomputed as $RRS = 100|S_{Vina,i,m}|/|S_{Vina,i,WT}|$ against the corrected LYS-76 wild-type denominator. K76T and K76A ratios span 99.1 to 100.6 percent and 99.4 to 100.4 percent respectively, and the pipeline-null ratios span 99.4 to 100.5 percent.

The null distribution therefore brackets the 80 %/70 % class boundaries entirely: the maximum deviation of any mutant ratio from the null ratio is 1.2 percentage points (K76T) and 0.5 percentage points (K76A), both far below the 10 % boundary spacing. The corrected PfCRT channel consequently carries no detectable K76T/K76A effect for this panel, and PfCRT RRS values near 100 percent are read as neutral-within-null rather than as evidence of active resistance circumvention. Class assignments, RRS means, and favorability counts in the main text reflect this corrected channel. The pipeline-null design follows the reviewer request (R2.5) for a batch-effect control: identical processing with only the substitution removed. The corrected PfCRT RRS values and class assignments appear in the main text (Table 1; Section 3.4). The neutral-within-null reading is independently corroborated by the companion external docking replication (39 ligands $\times$ 8 PfDHFR/PfCRT states, 312/312 finite records, prepared independently of the present panel), whose bootstrap RRS distribution of 100.45 [99.59, 101.32] brackets the corrected channel’s null spread; in the companion P2 layer, class assignments on its canonical panel are stable to bounded ± 1 kcal/mol score perturbations (unchanged in 212/272 records, 77.9%).

S13 Available machine-readable resources

The accompanying repository provides the computational record used in this study. The available resources include:

- the locked candidate manifest and canonical molecular identifiers;
- the complete target-wise Vina score matrix, ligand poses, and pose-provenance records;
- the declared wild-type/mutant docking panel used for the per-target RRS analysis;
- the derived RRS, PNS, ACSI, and cross-metric tables;
- the chemical-space similarity data and source data for the coverage figure;

- receptor and ligand input files, configuration files, analysis scripts, and figure-generation scripts;
- integrity manifests and reproducibility metadata.

Exact file names, directory paths, checksums, and version-specific dependencies are listed in the repository README and submission manifest rather than in the scientific narrative of this supplement. The corresponding evidence boundary and reproducibility statement in the main text is given in Section 2.6. These resources are intended to support independent computational reproduction; they do not convert docking scores into experimental affinity measurements.

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